

3 (**short**) The planar pin-jointed truss structure in Fig. 3 comprises an outer regular hexagon with all members of length L and two diagonal members QT and RU which **do not touch** each other. Node S is pinned to the ground, and node P cannot move horizontally. All members in the structure have the same cross-sectional area A and Young's Modulus E .

Derive an expression for the displacement of node P caused by the vertical load F as shown.

[10]

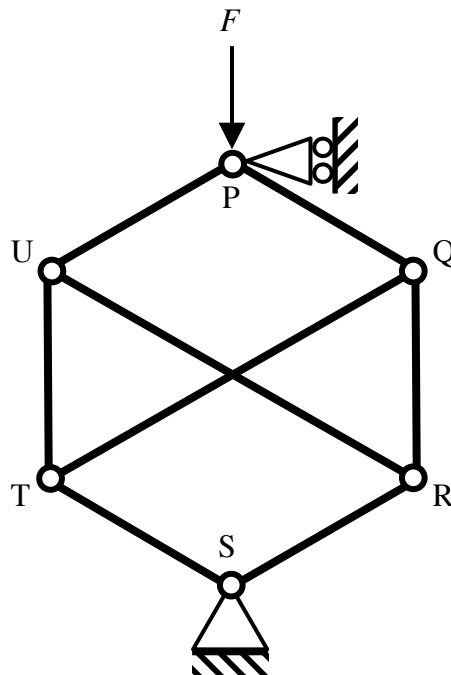


Fig. 3